NYC Road Safety Analysis: Understanding Collisions in 2020–2021

Dataset Description

For this project, we used the NYC Motor Vehicle Collisions dataset from NYC Open Data covering January 2020 to December 2021. The original file had over 223,000 rows. After cleaning, we worked with 141,727 usable records and 12 columns.

Key variables include:

- Crash Date & Crash Time
- Borough, Latitude, Longitude
- Contributing Factor
- Number of Persons Injured & Killed
- On Street Name & Cross Street Name
- Vehicle Type
- Our own calculated fields (explained below)

The original purpose of this data is to help the city track traffic accidents and improve road safety. It is a real world dataset that is messy, but interesting to analyze.

Data Preparation Narrative

The raw data had a lot of issues. For example, it has missing values in location fields, inconsistent street names, and too many columns we did not need.

We used a Jupyter notebook to clean it step by step:

- Filtered the data to only 2020–2021
- Removed rows missing Borough, Latitude, or Longitude (important for mapping)
- Filled in the one missing injured value with 0
- Kept only the most useful columns
- The dataset had five separate columns for contributing factors (Contributing Factor Vehicle 1 through 5), and most of them were null after the first column. We consolidated all the non-null values into a single normalized column so we could analyze causes more easily.
- We also removed rows that didn't have proper intersection information. Since our focus was on identifying dangerous intersections, we filtered out records that only had a single street name.

We also created three calculated fields in Tableau:

- Total Casualties = Injured + Killed
- Intersection Name = On Street Name + Cross Street Name
- Collision Category = grouped the Contributing Factor into five clear categories:
  - Human Error (Driver Inattention, Unsafe Speed, etc.)
  - Health (Alcohol, Drugs, Fatigue, Illness)

- Environmental (Slippery roads, Obstructions)
- Vehicle (Mechanical issues)
- Other/Unspecified

These steps made the data much cleaner and easier to visualize.

Design Rationale

We built two complementary dashboards that work together to tell one story:
1. What causes collisions?
2. Where do they happen?
3. How bad are the high risk spots?

Dashboard 1 – High-Level Overview
- Packed Bubble Chart (Primary Causes):
  - Great for showing proportions at a glance. Human Error is so big it immediately stands out.
- Horizontal Bar Chart (Collisions by Borough):
  - Simple and effective for comparing the five boroughs.
- Area Chart (Monthly Trends):
  - Perfect for showing the pandemic drop and recovery over time.
- Colors:
  - Bright orange for Human Error (the main problem), with consistent colors across both dashboards.

Dashboard 2 – Hotspots & High-Risk Intersections
- Density Map:
  - Shows exactly where collisions cluster in NYC.
- Stacked Bar Chart (Top 15 Intersections):
  - Shows both total casualties and the breakdown by category.
- Bar Chart (Total by Category):
  - Reinforces the main message from Dashboard 1.

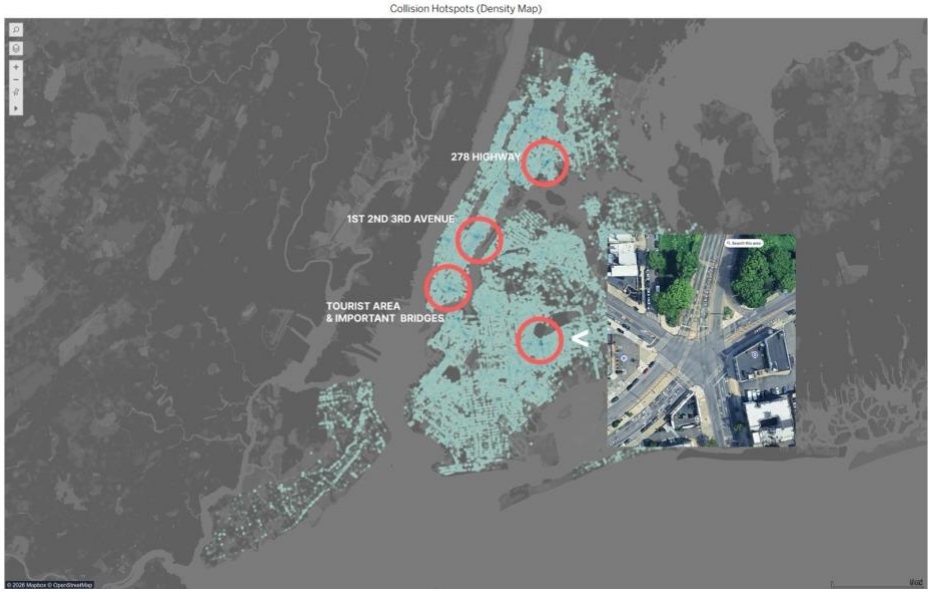
We used **Borough** and **Collision Category** filters on both dashboards so viewers can explore the data themselves. Clicking on charts also filters everything else (cross-filtering), which makes the dashboards feel interactive and useful.

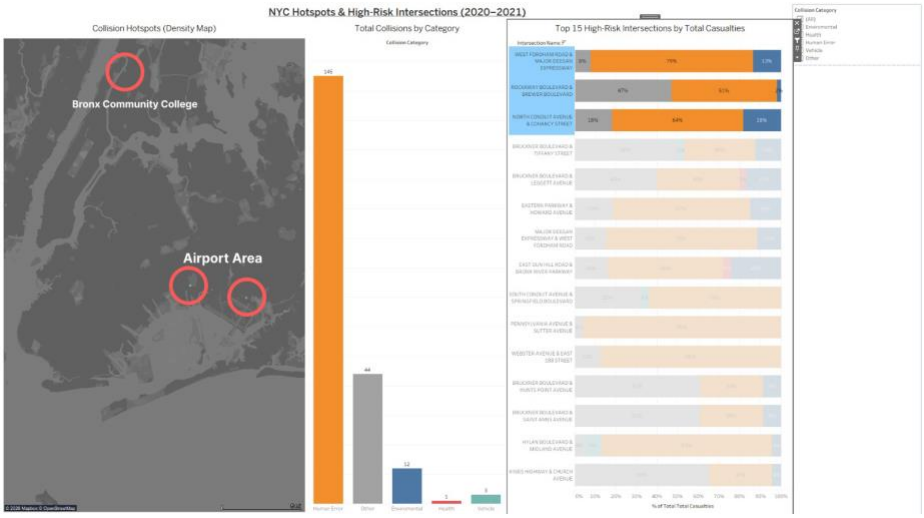
Analytical Insights

After exploring the dashboards, we found these key takeaways:
1. Human Error is the biggest problem:
   - It accounts for 66.7% of all collisions. This includes driver inattention, unsafe speed, failure to yield, etc.

2. Brooklyn and Queens have the most collisions (29K and 26K respectively), the worst individual intersections are spread across both boroughs.
3. The pandemic had a huge temporary effect:
    - Collisions dropped sharply in March/April 2020 (down to 1,500 per month), but started rising again as restrictions eased.
4. Certain intersections are especially dangerous:
    - Places like West Fordham Road & Major Deegan Expressway and Rockaway Boulevard & Brewer Boulevard had very high casualty numbers, mostly due to Human Error.
5. Even though total collisions dropped during lockdown, the proportion of Human Error crashes stayed consistently high, meaning behavior didn't really change.

While exploring the intersection data, we noticed that some of the most dangerous locations were complex intersections where 7 streets/ramps came together (like major expressway junctions). These spots had significantly higher casualty counts.





Limitations and Future Work

Limitations:

- Some records had missing street names, so our intersection analysis isn't 100% complete.
- We only looked at 2020–2021, so we don't know if patterns are different in other years.
- No data on weather, traffic volume, or speed limits, which probably affect crashes.

Future Work:

- Add weather data to see how rain/snow affects collisions.
- Include more years for better trend analysis.
- Maybe create a dashboard that shows injuries vs. fatalities specifically.

Overall, this was an interesting project. Tableau made it easy to turn messy city data into something clear and actionable. We learned a lot about data cleaning, calculated fields, and designing dashboards that tell a story.